

The topography of human mobility in pre-modern Europe

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Abstract

Human mobility has long been shaped by the constraints of terrain, yet the large-scale geography of mobility in (pre-)history remains poorly understood. Considering slope gradient as the primary influence on mobility, this study introduces a spatial choice model to estimate mobility preference from documented pre-modern routes in northern and central Europe. Mobility preference extrapolated to the continental scale reproduces mobility corridors along valleys, coastal plains, and gently sloping gradients, highlights mobility barriers such as the Alps, and predicts undocumented routes between pre-modern settlements. We further assess predicted mobility preference at documented pre-modern settlements and find a significant association. The model presented here provides a replicable method for estimating mobility preference from heterogeneous archaeological traces of movement, enabling new perspectives on how mobility structured patterns of connectivity in human history.

Introduction

Human mobility is the invisible infrastructure of (pre-)history. Where people chose to move structured the circulation of goods, the campaigns of armies, the journeys of pilgrims, and the diffusion of both ideas and diseases, shaping the economic, political, and cultural development of a region, including pre-modern Europe¹⁻⁴. Yet the mobility preference of humans in (pre-)history, materialised as routes (tracks and roads) in the landscape, remains unclear, hindered by fragmentary historical records, heterogeneous archaeological traces, and local case studies that resist continent scale synthesis^{2,5}. At the general level, empirical evidence shows that tracks in pre-modern (late medieval and early modern, 1350-1650AD) Europe were not formally engineered but emerged through repeated use, often as multiple parallel tracks across valleys or along shallow slope gradients^{2,4}. Maintenance was minimal and usually the responsibility of local communities, leaving tracks vulnerable to deterioration, seasonal impassability, and frequent re-routing.

Archaeologists and historical geographers have long used least-cost path (LCP) analysis to model (pre-)historic movement⁶⁻⁸. The LCP approach, however, fails to explicitly model the underlying mobility preference that structured movement, and which ultimately produced the route

systems observed in the archaeological and historical record. Here, I introduce a slope-based spatial choice model (henceforth SCM) to estimate mobility preference from documented late medieval and early modern routes (1350-165) in northern and central Europe^{9,10}. The empirically grounded pre-modern mobility preference is then used to predict mobility preference across both the wider region of northern and central Europe and the continent of Europe as a whole. Predicted maps of mobility preference at both the regional and continental scale are statistically validated using historical evidence of pre-modern settlements, which serve as an independent proxy for locations where human movement would have occurred.

This SCM approach provides a quantitative framework for inferring mobility preference from fragmentary archaeological and historical records while explicitly linking human spatial choices to the emergence of route systems. Results show that a parsimonious slope-based SCM reproduces key features of documented pre-modern route systems. Crucially, the SCM also predicts plausible mobility corridors at different scales, including areas where archaeological and documentary evidence is limited or unavailable.

Modelling mobility preference from fragmentary records

The Spatial choice model (SCM) provides an explicit, statistical framework to estimate mobility preference from documented (pre-)historic routes (**see Materials and Methods**). Whereas LCP analysis infers preference indirectly by comparing a predicted optimal path to a documented route¹¹, the SCM explicitly evaluates each segment of movement along a documented route as an individual choice situated within its local landscape context. For each movement segment, a set of spatially defined alternatives is constructed that reflects the possible movement an individual could have taken from that point in the landscape. The observed choice among these alternatives is used to infer mobility preference. While numerous social, political, and environmental factors may have shaped mobility preference at a local scale, the SCM focuses on mathematical slope gradient as a consistently measurable variable across both regional and continental scales. By incorporating mathematical slope gradient as the primary explanatory factor, the SCM quantifies how variation in slope gradient influences the selection of one alternative over another. Leveraging multiple routes, the SCM yields a single, generalisable function describing mobility preference in relation to slope gradient in pre-modern Europe. Regional and continental scale predicted maps of mobility preference were compared spatially with the location of 1,247 regional and 2,180 continental pre-modern settlements (**Extended Data Fig. 1-2**). As these settlements were not inputs in the SCM, they serve as independent proxies to compare predicted mobility preference to where movement would have occurred. To statistically assess this relationship, we compared predicted mean mobility

preference for pre-modern settlements to equal numbers of randomly simulated points (999 iterations) at both a regional and continental scale (see **Materials and Methods**).

Results

The slope-based spatial choice model estimates that mobility preference in pre-modern Europe declines sharply as slope gradients become steeper ($b = 14.995$; $SE = 0.087$; $p < 0.001$) (**Fig. 1**). At a gradient of 0.05 (2.86°), mobility preference reduces by more than half (52.75%), and by 0.16 (9.09°) it has declined by over 90%. For comparison, relative speed while hiking on foot¹² reduces by only 9.97% and 42.88% at 0.05 and 0.16 slope gradient, respectively.

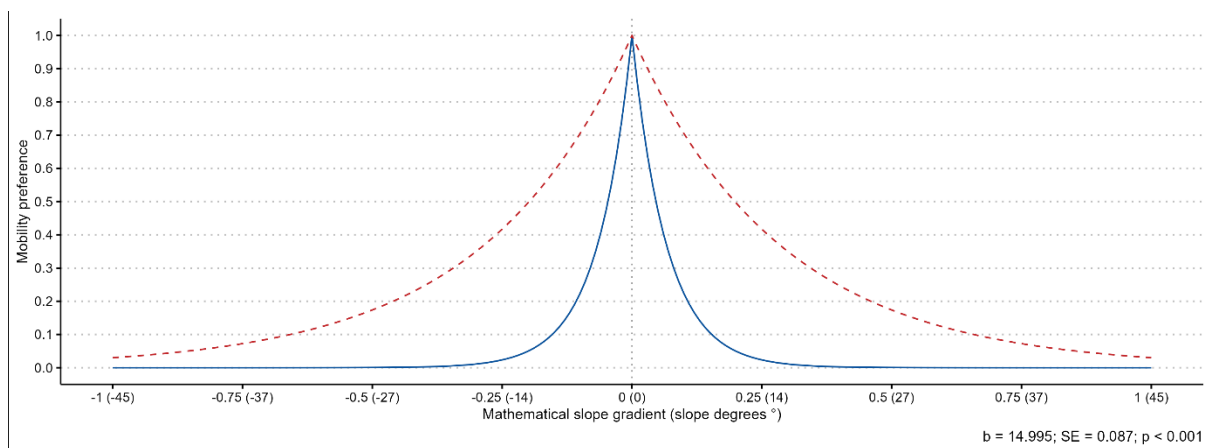


Figure 1. Predicted mobility preference from documented late medieval and early modern routes (1350-1650AD) in northern and central Europe (blue line). Predicted mobility preference while hiking on foot¹² (red dashed line)

The spatial choice model reveals the continental scale geography of mobility preference in pre-modern Europe (**Fig. 2**). Regions of high mobility preference are concentrated along valleys, coastal plains, and gently sloping gradients, while steep mountain ranges and other topographically challenging areas show low preference. The Alps, for example, emerge as a major barrier to mobility, punctuated only by narrow high-preference passages that align with historically significant trans-alpine routes¹³ (**Extended Data Fig. 3**).

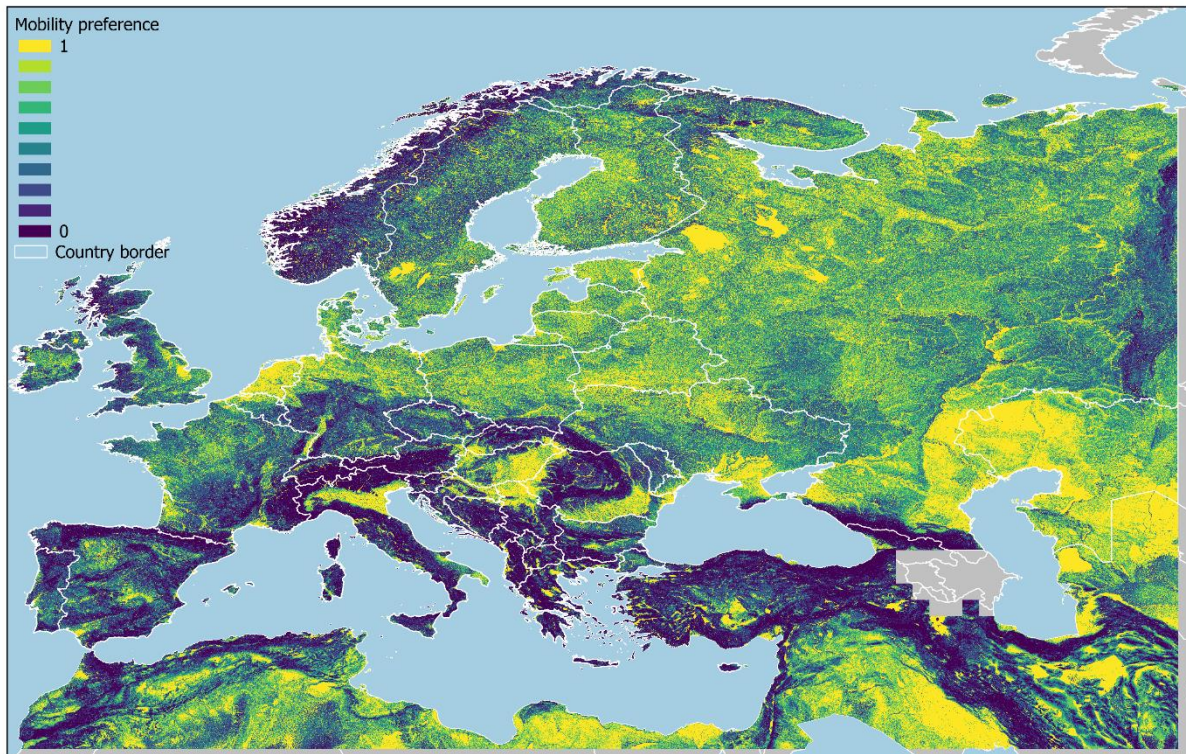


Figure 2. Predicted mobility preference in pre-modern Europe (late medieval and early modern, 1350-1650AD)

Out of the 1,247 regional and 2,180 continental pre-modern settlements, 1,062 (85.16%) and 1,388 (63.67%) have a predicted mobility preference greater than 0.5. Pre-modern settlements at the regional scale show a more concentrated distribution towards higher mobility preference compared to the broader distribution observed at the continental scale (**Extended Data Fig. 4**). Pre-modern settlements at the regional scale have a higher mean predicted mobility preference (0.750) than settlements at the continental scale (0.599) (**Fig. 3**). At both scales, predicted values for settlements are significantly higher than those from 999 iterations of randomly simulated points ($p < 0.001$; **See Methods and Supplementary Information**).

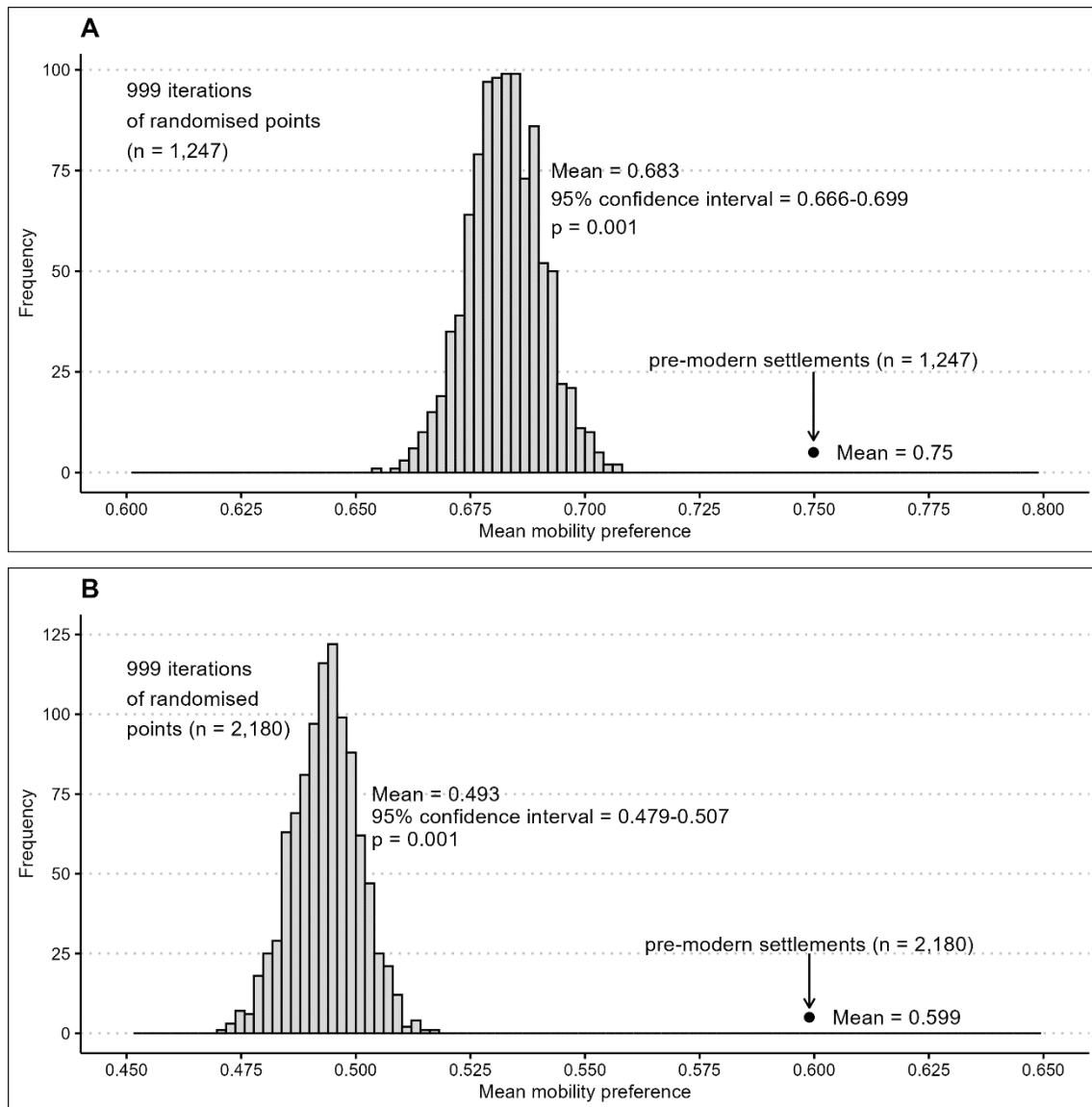


Figure 3. Predicted mean mobility preference extracted from 999 iterations of randomly simulated points (left histograms) and predicted mean mobility preference for pre-modern settlements (arrows). Mobility preference values extracted at the regional scale (mean mobility preference of random points = 0.683 versus mean mobility preference of pre-modern settlements = 0.75, $p < 0.001$) (A). Mobility preference values extracted at continental scale (mean mobility preference of random points = 0.493 versus mean mobility preference of pre-modern settlements = 0.599, $p < 0.001$) (B).

Discussion

The spatial choice model shows that slope gradient was a dominant constraint on mobility preference in pre-modern Europe, with even modest slopes dramatically reducing preference. This aversion to steeper slope gradients channelled potential movement in pre-modern Europe along

lowland corridors, valleys, and gentle slope gradients. The distributions of predicted mobility preference for pre-modern settlements (**Extended Data Fig. 4**) suggests that mobility preference was more consistently expressed at the regional (northern and central Europe) scale, where the slope-based spatial choice model was fitted. The broader distribution of predicted mobility preference for pre-modern settlements at the continental scale reflects greater spatial and contextual variability at larger spatial scales. That pre-modern settlements at both regional and continental scales, however, differ significantly from randomly simulated points indicates that the model is not simply overfitting to the regional scale but is capturing structural properties of mobility preference that generalises from the regional scale to the continental scale.

By quantifying how slope gradient structured mobility preference and shaped local choices for where people might have moved, the high-resolution maps of mobility preference in pre-modern Europe reproduce large-scale mobility corridors while simultaneously highlighting topographic barriers to mobility. Predicted mobility preference from the SCM also reveals smaller-scale routes connecting pre-modern settlements that are not documented in historical sources⁹, such as from Opole to Brzeg and Nysa, Poland (**Extended Data Fig. 5a**). With archaeological and documentary evidence of pre-modern routes often poor^{2,4,5}, the slope-based spatial choice model provides a quantitative approach to identify plausible routes where direct evidence is lacking (**Extended Data Fig. 5b, c**).

This study opens several avenues for future research: it establishes a quantitative baseline of mobility preference in pre-modern Europe that enables systematic comparison across regions and periods; produces maps of predicted mobility preference that form testable hypotheses, e.g. how mobility preference influenced trade, communication, and settlement development; and offers a methodological alternative to standard least-cost path analyses that emphasises where people chose to move over assumptions of optimality.

Materials and Methods

Overview of the slope-based spatial choice model

The spatial choice model (SCM) provides an explicit, statistical framework to estimate mobility preference from documented (pre-)historic routes. With the explicit evaluation of each segment of movement along a documented route as an individual choice event situated within its local landscape context, the mobility preference of individuals choosing a specific movement among possible alternatives can be estimated. Here, the primary explanatory factor on mobility preference is slope gradient but additional factors can be incorporated within the SCM framework.

A single movement along a documented route is modelled as a Poisson realisation. For each segment of movement—an individual choice event—individuals can choose between eight adjacent directions from their current location: North (N), Northeast (NE), East (E), Southeast (SE), South (S), Southwest (SW), West (W), Northwest (NW). The chosen direction for movement among these eight alternative movement choices is taken to reflect the mobility preference of individuals in pre-modern Europe. Collectively, over multiple single movements and documented routes, the SCM yields a single, generalisable function describing mobility preference.

Following¹², the influence of mathematical slope gradient is modelled using an isotropic, double exponential (Laplace) function:

$$M = e^{-c|s|}$$

where M is mobility preference; $-c$ controls how fast mobility preference decays with slope gradient; and s is the absolute slope gradient (elevation difference/horizontal distance)

This exponential decay form is equivalent to expressing the linear predictor of each alternative's relative mobility preference as a linear function of negative slope within a Poisson framework:

$$\eta = c(-s)$$

where $c = \beta$ is the estimated coefficient representing the influence of slope gradient on mobility preference

The choice probability is thus proportional to the exponent of the linear predictor, or:

$$P(\text{relative mobility preference}) \propto e(\eta) = e^{-c|s|}$$

Base datasets

Documented late medieval and early modern routes (1350-1650AD) and pre-modern settlements ($n = 1,247$) in northern and central Europe were obtained from the *Viabundus Project*^{9,10} (**Extended Data Fig. 6**). To minimise the influence of modern-day roads when estimating pre-modern mobility preference, route sections located within 1km of a modern road were excluded from the analysis, reducing the total length of pre-modern routes from 79,645km to 6,792km (**Extended Data Fig. 7**). Modern roads (highways, primary roads, secondary roads, and tertiary roads, but excluding local roads) were derived from *Global Roads Inventory Project (GRIP)* dataset¹⁴. Pre-modern settlements at the continental scale were obtained from *The Population of European Cities from 700 to 2000* dataset¹⁵ (**Extended Data Fig. 2**). The ~30m Forest And Buildings removed Copernicus Digital Elevation Model (*FABDEM*)¹⁶ was used to calculate mathematical slope gradients for both regional and continental scale analyses. Note that the modern countries of Armenia and Azerbaijan were not included in FABDEM due to country restrictions.

Calculating slope gradient along documented routes

For each segment of movement along routes, mathematical slope gradients (elevation difference divided by horizontal distance) were calculated from each focal cell that is crossed by a route to its eight adjacent cells. Routes were defined such that each route begins and ends either 1) at the intersection with another documented route, or 2) at the point where the route intersects the 1 km buffer around a modern road. In total, 2,856,864 mathematical slope gradients (1,929 routes and 357,108 segments of movement) were modelled to estimate mobility preference in pre-modern Europe.

Estimating mobility preference from segments of movement

The mobility preference of individuals in pre-modern Europe was estimated using a Poisson generalised linear mixed model (GLMM) using the R package *lme4*¹⁷ and the *glmer* function. The spatial choice model treats each movement alternative as an observation grouped within a set of possible movement alternatives (i.e. each movement segment) and includes segment-specific offsets to account for the multi-level structure:

$$P(\text{relative mobility preference}_{\{ij\}}) \propto e(\eta_{\{ij\}}) = e(\beta \cdot (-s_{\{ij\}}) + u_i)$$

where $\eta_{\{ij\}}$ is the linear predictor representing the relative mobility preference of alternative i in movement segment j ; $-s_{\{ij\}}$ is the negative slope gradient of that alternative; β is the estimated coefficient representing the influence of slope gradient on relative mobility preference; and u_i is the

random intercept for movement segment j , capturing baseline differences among movement segments.

Predicting mobility preference at the regional and continental scale

To obtain predicted values of mobility preference from the fitted *glmer* spatial choice model, conditional predictions of mobility preference as a function of slope gradient at increments of 0.01 between -1 to 0 were calculated using the R package *marginaleffects*¹⁸ (**Supplementary Data 1**). Predictions were based on fixed effects only (i.e. excluding random intercepts; *re.form = NA*) to estimate a generalisable, population-average mobility preference in pre-modern Europe. The predicted mobility preference values were applied to maps of slope gradients at both the regional and continental scale to produce continuous surfaces of mobility preference in pre-modern Europe.

Pre-modern settlements versus randomly simulated points

To quantify the spatial association between documented pre-modern settlements as independent proxies for where movement would have occurred and maps of predicted mobility preference at both regional and continental scales, I generated 999 unique iterations of randomly simulated points equal to the number of pre-modern settlements at each scale (regional, $n = 1,247$; continental, $n = 2,180$). To account for spatial heterogeneity in documented pre-modern settlements, with some areas having more settlements than others, the randomly simulated points were stratified by modern country borders so that the number of random points was equal to the number of documented pre-modern settlements in each country.

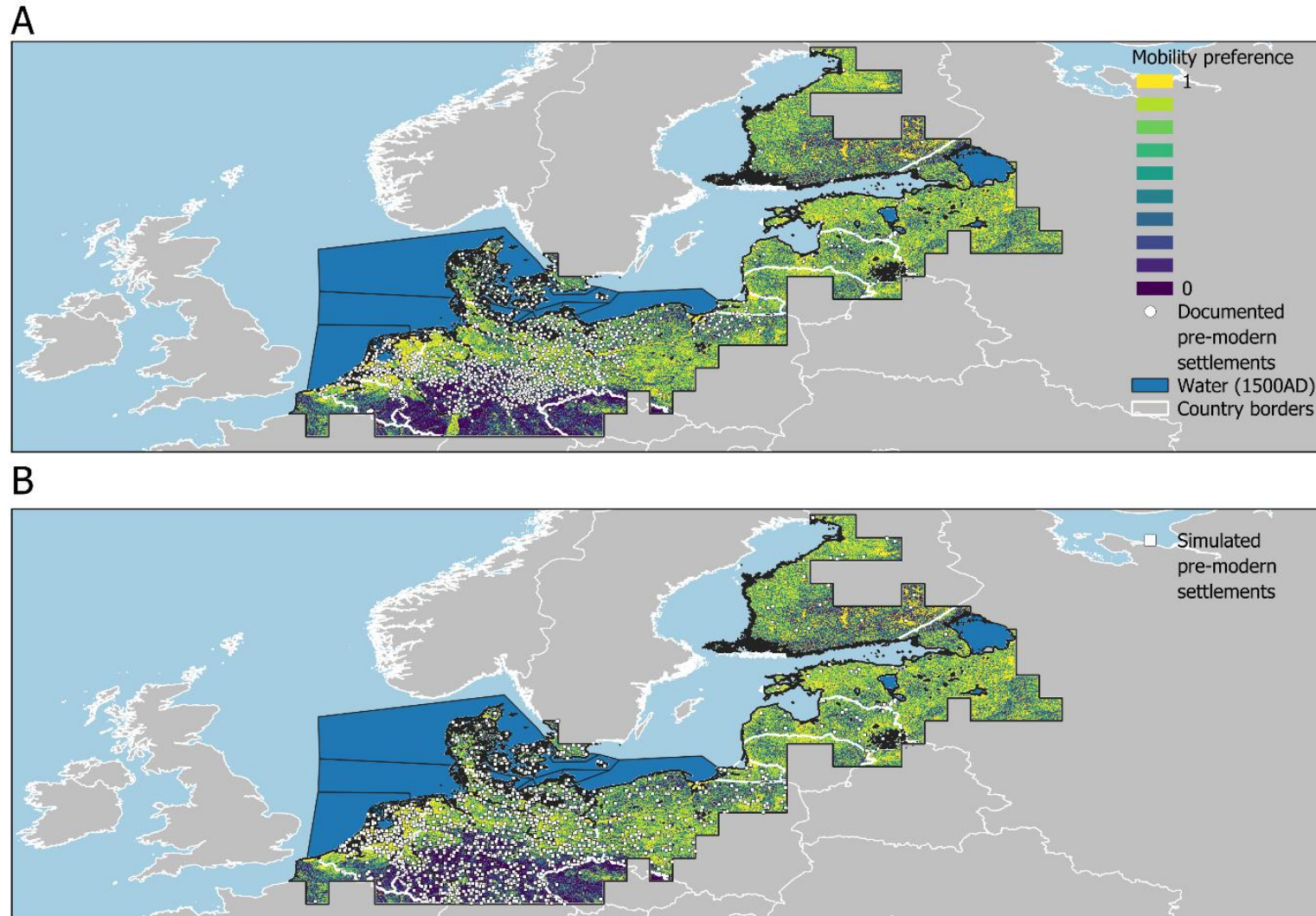
To statistically assess spatial association, predicted mobility preference values were extracted for pre-modern settlements at both the regional and continental scale. This process of extraction was repeated for the 999 iterations of randomly simulated points at both scales. Using these data, the mean predicted mobility preference values for pre-modern settlements at the regional and continental scale was compared against 999 mean predicted mobility preference values for the 999 iterations of randomly simulated points (the null hypothesis). The exact *P value* (the probability of obtaining a mean predicted mobility preference equal to or greater than the observed mean predicted mobility preference under the null) was calculated via a Monte Carlo Hypothesis Test¹⁹: $(r + 1)/(n + 1)$, where r = the number of replicates that produce a test statistic greater than or equal to that calculated for the observed data, and n = number of replicate samples that have been simulated.

References

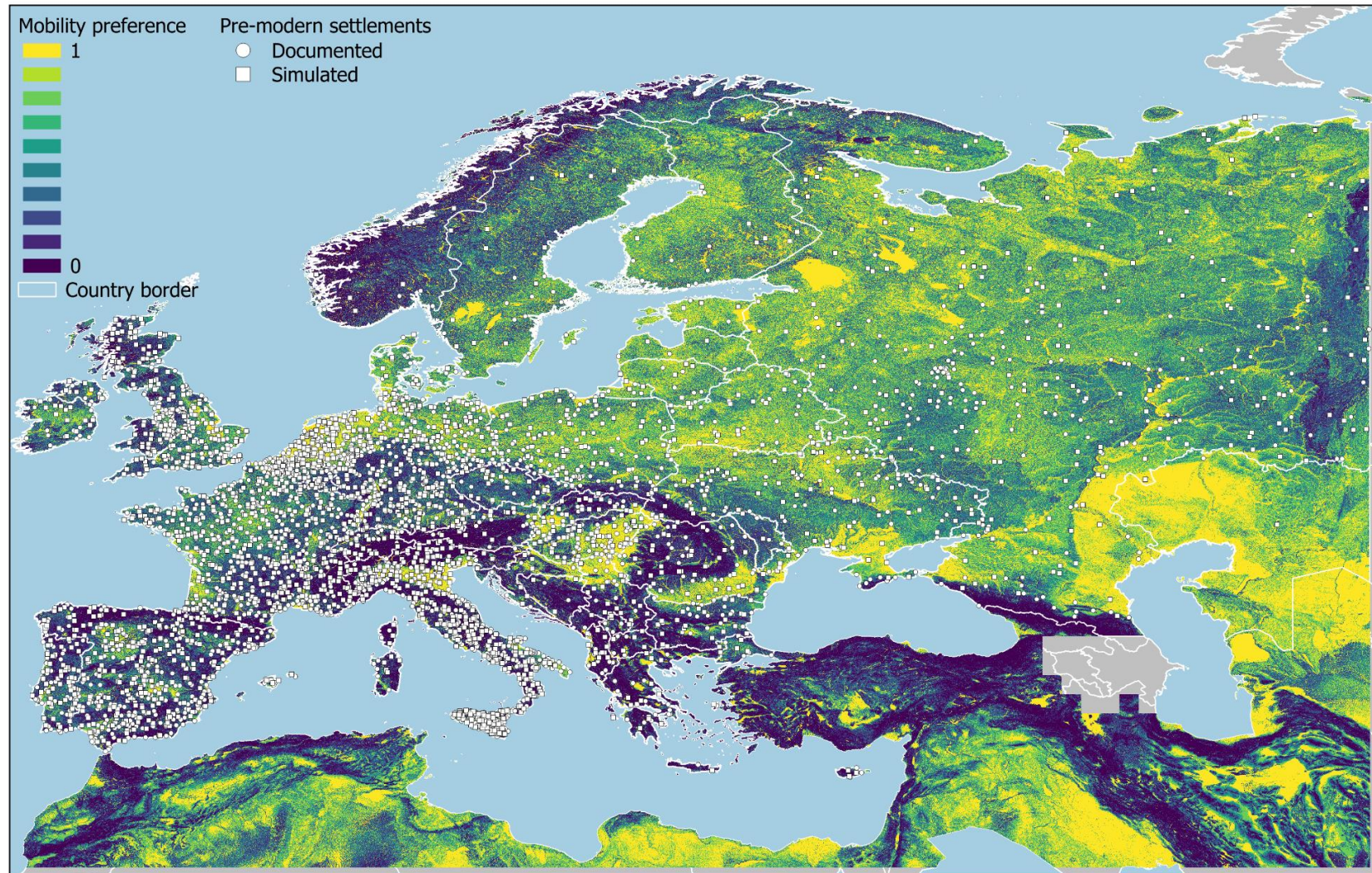
1. Ohler, N. *The Medieval Traveller*. (The Boydell Press, Woodbridge, Suffolk, 1989).
2. Hindle, B. P. *Medieval Roads and Tracks*. (Shire, Princes Risborough, 1998).
3. Webb, D. *Pilgrims and Pilgrimage in the Medieval West*. (I.B. Tauris, London, 2001).
4. Scott, H. M. Travel and Communications. in *The Oxford handbook of early modern European History, 1350-1750* (ed. Scott, H. M.) 165–191 (Oxford University Press, Oxford, 2015).
5. Hindle, B. P. Sources for the English medieval road system. in *Roadworks: medieval Britain, medieval roads* (eds. Allen, V. & Evans, R.) 33–49 (Manchester University Press, Manchester, 2016).
6. Lewis, J. Probabilistic Modelling for Incorporating Uncertainty in Least Cost Path Results: a Postdictive Roman Road Case Study. *J Archaeol Method Theory* **28**, 911–924 (2021).
7. Kabadayi, M. E., Gerrits, P. & Boykov, G. Geospatial mapping of a 16th century transport corridor for Southeast Europe. *Digital Scholarship in the Humanities* **37**, 788–812 (2022).
8. Herzog, I. 11 Reconstructing past movement patterns on the basis of known old roads. A case study on pre-industrial roads in a hilly area in Germany. in *Mobility in the Early Middle Ages, and Beyond – Mobilität im Frühmittelalter und darüber hinaus* (eds. Sarti, L. & Von Trott Zu Solz, H.) 213–242 (De Gruyter, 2025). doi:10.1515/9783111166698-011.
9. Holterman, B., A.B., M., Andersen, K. H., Dengg, M. C. & Petersen, N. Viabundus: Map of Premodern European Transport and Mobility: Social and Economic History. *Res. Data J Humanit. Soc. Sci.* **7**, 1–13 (2022).
10. Holterman, H. Viabundus Pre-modern Street Map 1.3. <https://www.viabundus.eu> (2024).
11. Lewis, J. Explaining Known Past Routes, Underdetermination, and the Use of Multiple Cost Functions. *J Archaeol Method Theory* (2023) doi:10.1007/s10816-023-09621-w.
12. Tobler, W. *Three Presentations on Geographical Analysis and Modeling. Technical Report 93-1*. <https://escholarship.org/uc/item/05r820mz> (1993).

13. Spufford, P. *Power and Profit: The Merchant in Medieval Europe*. (Thames & Hudson, London, 2006).
14. Meijer, J. R., Huijbregts, M. A. J., Schotten, C. G. J. & Schipper, A. M. Global patterns of current and future road infrastructure. *Environmental Research Letters* (2018).
15. Buringh, E. The Population of European Cities from 700 to 2000: Social and Economic History. *Res. Data J Humanit. Soc. Sci.* **6**, 1–18 (2021).
16. Hawker, L. *et al.* A 30 m global map of elevation with forests and buildings removed. *Environ. Res. Lett.* **17**, 024016 (2022).
17. Bates, D., Mächler, M., Bolker, B. & Walker, S. Fitting Linear Mixed-Effects Models Using lme4. *Journal of Statistical Software* **67**, 1–48 (2015).
18. Arel-Bundock, V., Greifer, N. & Heiss, A. How to Interpret Statistical Models Using margineffects for R and Python. *Journal of Statistical Software* **111**, 1–32 (2024).
19. North, B. V., Curtis, D. & Sham, P. C. A Note on the Calculation of Empirical P Values from Monte Carlo Procedures. *The American Journal of Human Genetics* **71**, 439–441 (2002).

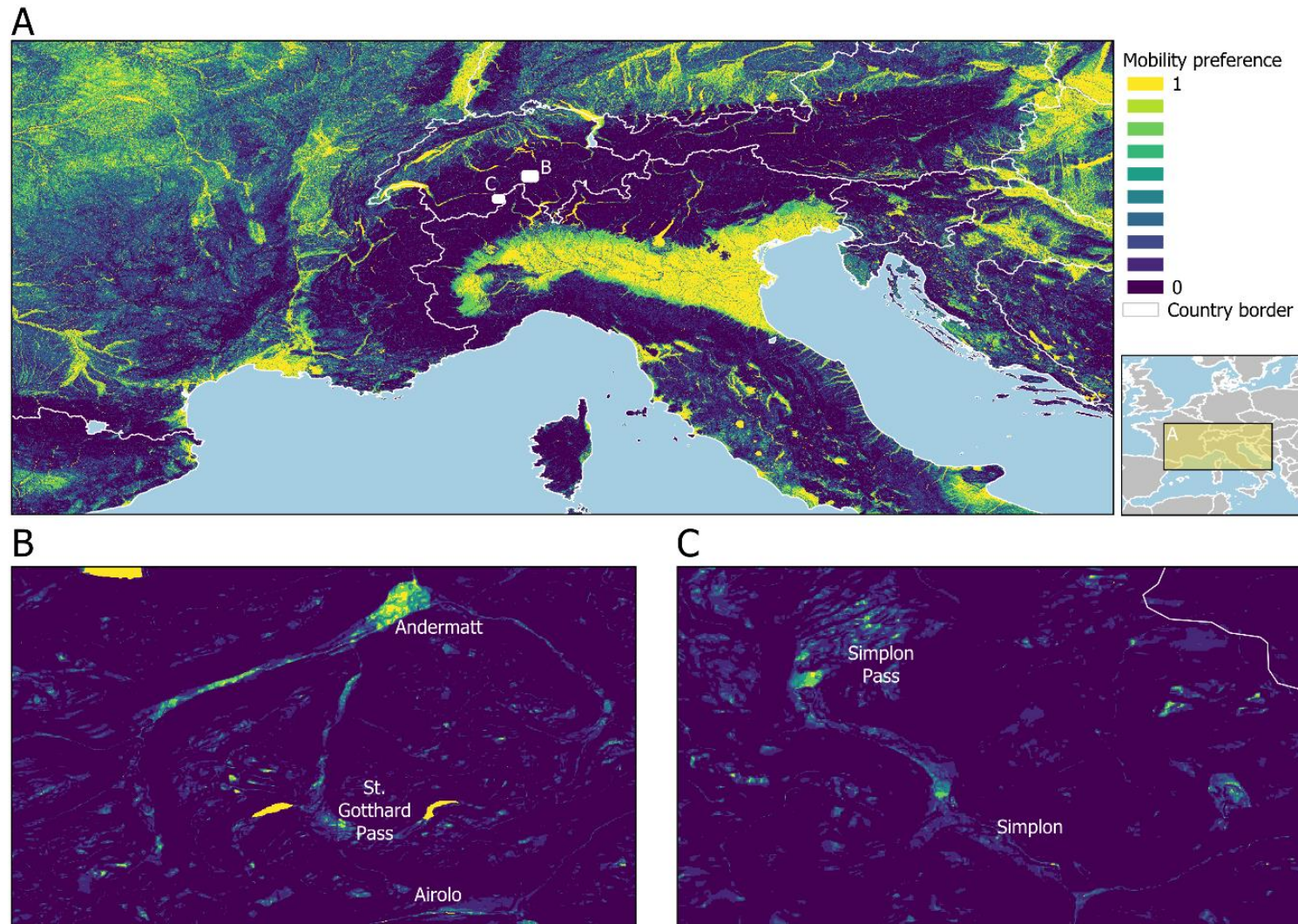
Extended Data Figures



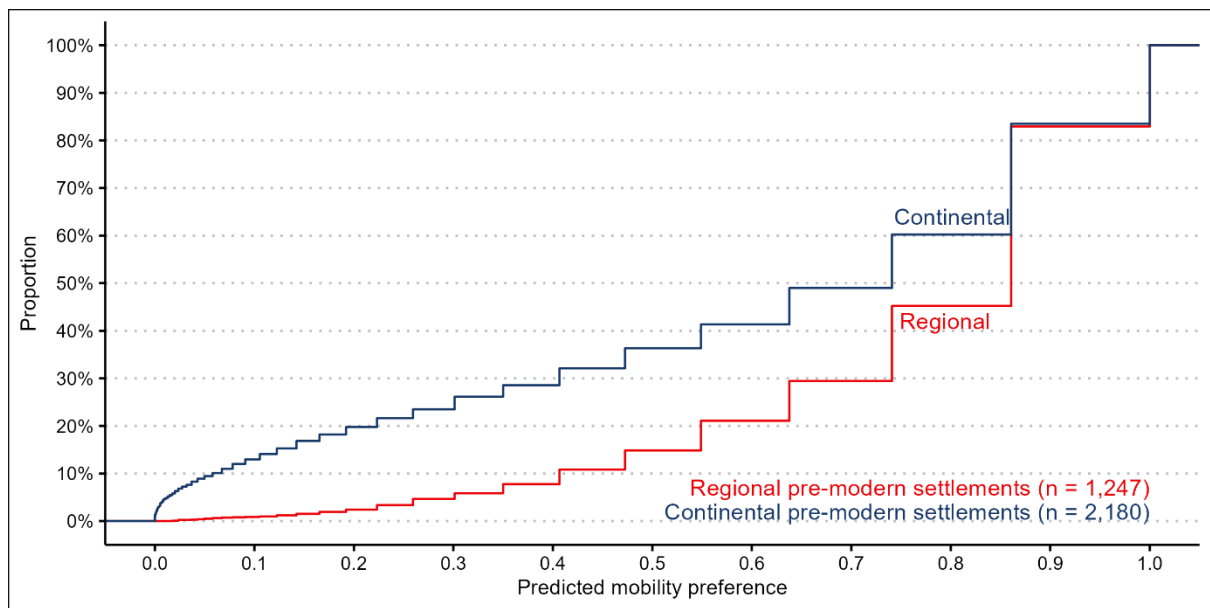
Extended Data Figure 1. Predicted mobility preference for northern and central Europe. Documented pre-modern settlements⁹ ($n = 1,247$) (**A**) and a single iteration of simulated pre-modern settlements (**B**)



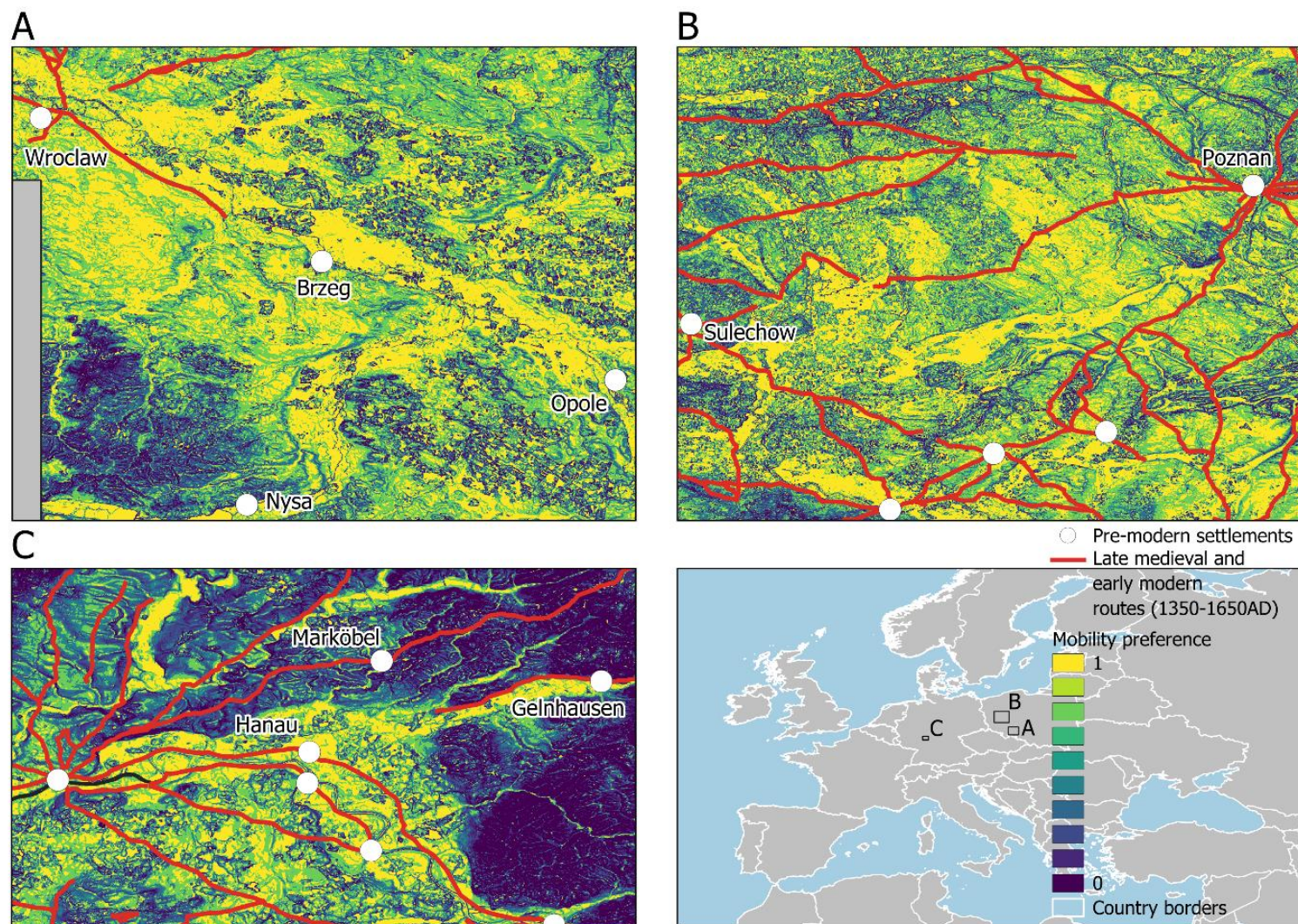
Extended Data Figure 2. Predicted mobility preference in pre-modern Europe. Documented pre-modern settlements¹⁵ and a single iteration of simulated pre-modern settlements (square and circle, respectively) across continental Europe (n = 2,180)



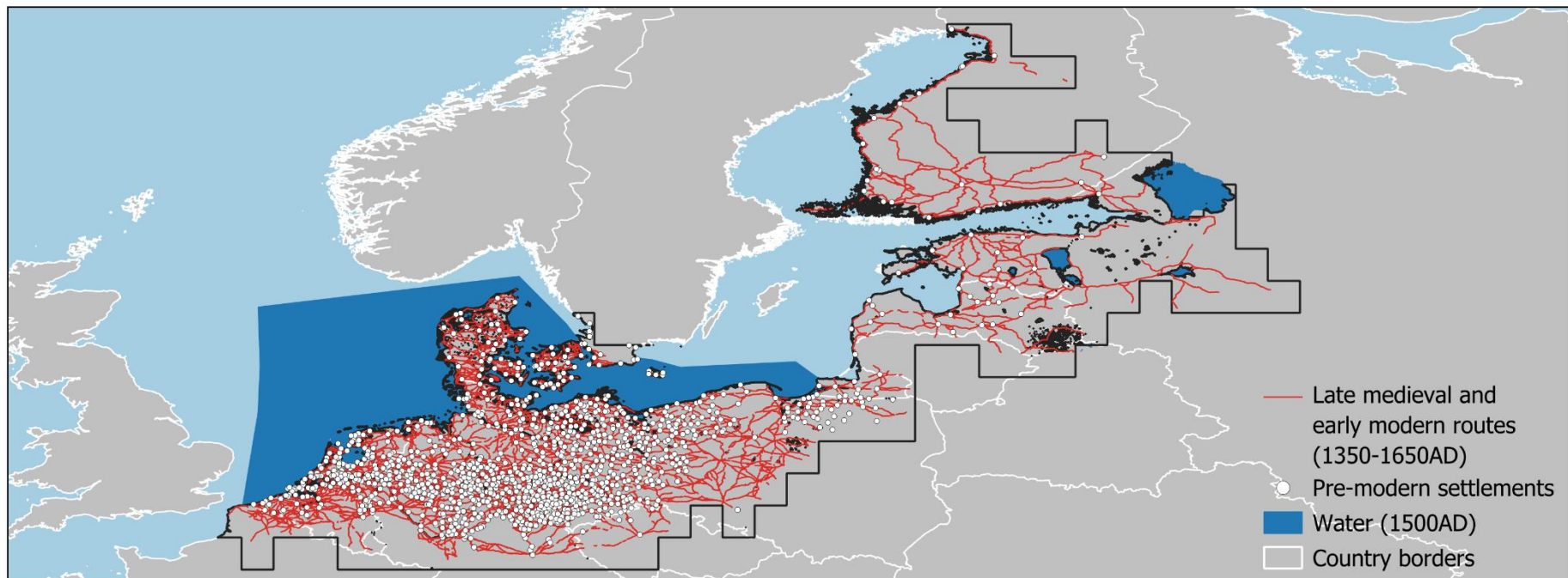
Extended Data Figure 3. Predicted mobility preference in pre-modern European Alps (**A**). Predicted mobility preference for the thirteenth century St. Gotthard Pass from Airolo to Andermatt, Switzerland (**B**) and Simplon pass, Switzerland (**C**)¹³



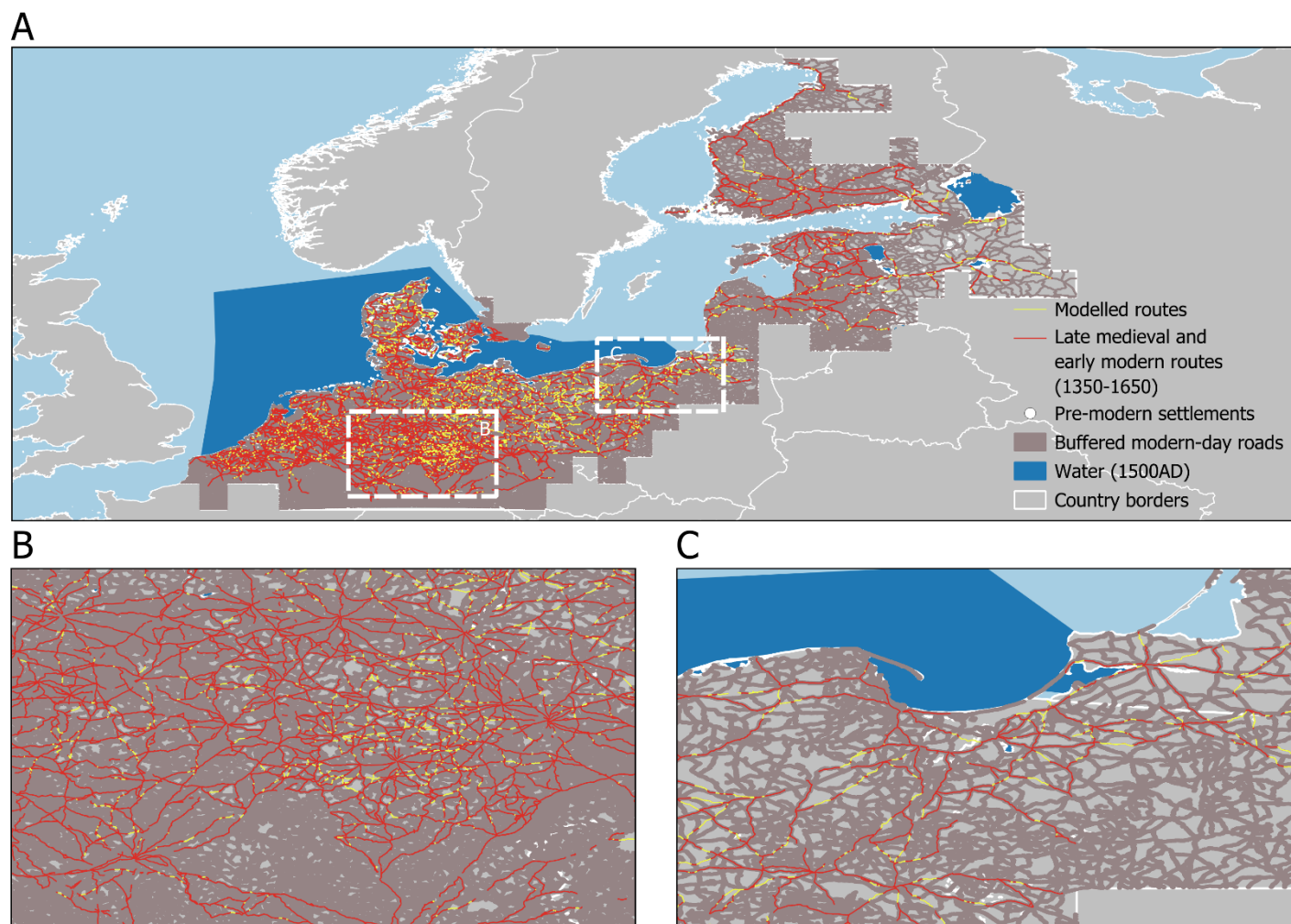
Extended Data Figure 4. Empirical cumulative distribution function of predicted mobility preference for pre-modern settlements at the regional ($n = 1,247$) and continental scale ($n = 2,180$)



Extended Data Figure 5. Proposed routes between pre-modern settlements from predicted mobility preference in pre-modern Europe. Proposed route corridors between Brzeg/Nysa and Opole, Poland (**A**), Poznań and Sulechów, Poland (**B**), and Gelnhausen/Marköbel and Hanau, Germany (**C**)



Extended Data Figure 6. Documented late medieval and early modern routes (1350-1650AD) (79,645km) and pre-modern settlements ($n = 1,247$) in northern and central Europe⁹



Extended Data Figure 7. Documented late medieval and early modern routes (1350-1650AD)⁹ inside 1km buffer from modern-day roads (red, 72,853km) and outside 1km buffer from modern-day roads (yellow, 6,792km) (**A**). Details of late medieval and early modern routes in modern-day Germany (**B**) and modern-day Poland (**C**)